

**D6 Modelling with
sequences A level only**

Name: _____

Class: _____

Date: _____

Time: **8 minutes**

Marks: **7 marks**

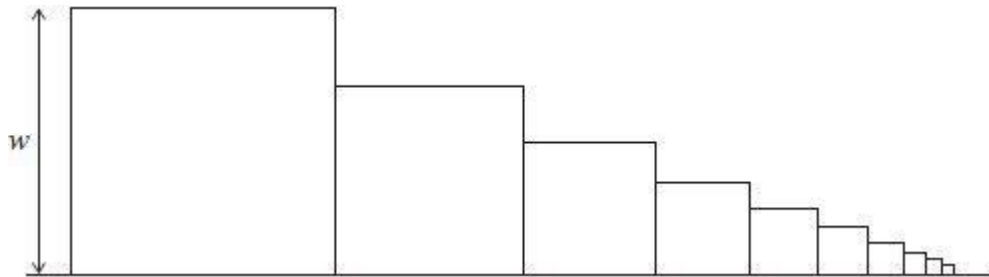
Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

Helen is creating a mosaic pattern by placing square tiles next to each other along a straight line.



The area of each tile is half the area of the previous tile, and the sides of the largest tile have length w centimetres.

- (a) Find, in terms of w , the length of the sides of the second largest tile.

(1)

- (b) Assume the tiles are in contact with adjacent tiles, but do not overlap.

Show that, no matter how many tiles are in the pattern, the total length of the series of tiles will be less than $3.5w$.

(4)

- (c) Helen decides the pattern will look better if she leaves a 3 millimetre gap between adjacent tiles.

Explain how you could refine the model used in part (b) to account for the 3 millimetre gap, and state how the total length of the series of tiles will be affected.

(2)

(Total 7 marks)

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.